

Measuring Austerity Policies in the Context of Crises: Evidence from an LP–DiD Framework with Debt Thresholds

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Abstract

This paper evaluates the relationship between high public and private debt and macroeconomic performance through the local-projection difference-in-differences (LP-DiD) framework. Drawing on macroeconomic data from advanced economies for 1972–2020, this paper estimates the dynamic responses of macroeconomic conditions to high public and private debt threshold crossings. This paper also contextualizes these findings with financial crises to test for endogeneity and the severity of macroeconomic conditions. This paper finds that high public debt thresholds, such as the benchmark 90% ratio, coincide closely with financial crises and are associated with already declining Real GDP growth rates. In contrast, high private debt thresholds predict sharper and more persistent contractions in GDP, consumption, and employment.

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1. Introduction

Following the 2008 Financial Crisis, governments sought optimal fiscal and financial policy responses to stabilize economies. Early academic work proved influential; notably, Reinhart and Rogoff (2009) argued that public debt-to-GDP ratios above 90

This research conflict mirrors the policy disagreements resurfacing today. In 2025, the United States began major cuts to international aid, limits on public health spending, and the potential privatization of student loans and housing finance. Meanwhile, Germany is reconsidering modifications to its debt brake rule to enable investment-led recovery. Against this backdrop, a critical question arises: When countries cross high public-debt thresholds after financial crises, do they truly experience lower GDP growth, or is the debt expansion part of a successful recovery process?

This paper addresses this question using an empirical framework tracing the response of macroeconomic outcomes—Real GDP growth, consumption, unemployment, and investment—around discrete debt-to-GDP thresholds. Using the MacroFinance MacroHistory Database (Jordà et al., 2017, 2021a), I focus on advanced economies from 1970–2020 and estimate local projection difference-in-differences (LP-DiD) impulse responses to public and private debt thresholds. The study examines a 90

The analysis proceeds in three steps. First, I estimate LP-DiD impulse responses using a historical panel dataset of countries' macroeconomic and financial controls, controlling for country- and year-fixed effects. Second, I examine the "negative horizons" of the LP-DiD model to assess pre-treatment dynamics, analogous to event-study checks in difference-in-differences, to evaluate endogeneity in the timing of debt accumulation. Third, I re-estimate the model, restricting the sample to crossings within three years of a financial crisis, to directly compare with the full-sample estimates.

The results in this paper reveal evidence of endogenous timing of the debt threshold. In the full sample, high public debt thresholds appear to be followed by weaker GDP perfor-

mance. However, negative horizons indicate that Real GDP and consumption growth had already been declining several years before public debt reached 90% of GDP. Once crisis timing is accounted for, post-threshold effects become smaller and less precisely estimated. In contrast, high private debt thresholds are associated with deeper and more persistent declines in output and consumption, which is consistent with the idea that private-sector deleveraging plays a central role in post-crisis adjustment ([Mian and Sufi, 2010](#)).

These findings contribute to the literature on debt cycles and economic growth. Treating high public debt as a direct cause of slow growth, rather than as a response to weak economic conditions, risks misattribution. A more nuanced perspective emphasizes sequencing of private and public balance-sheet adjustments and highlights circumstances under which fiscal consolidation may impede, rather than support, recovery ([Jordà et al., 2013](#)).

2. Literature Review

This paper aims to connect foundational literature on public debt cycles with recent research on credit cycles, employing recent advances in local projections with staggered treatment timing (LP-DiD). Foundational research focused on whether high public debt constrains future growth, with [Reinhart and Rogoff \(2010\)](#) arguing for sharp growth slowdowns above particular thresholds. Later studies [Panizza and Presbitero \(2014\)](#) indicated that public debt often rises in response to recessions, complicating causal inference from cross-country panels.

A related line of research studies macro-level credit cycles and reframes growth dynamics through the evolution of private debt rather than public debt alone. Work based on the JST MacroFinance & MacroHistory Database ([Schularick and Taylor, 2012](#); [Jordà et al., 2013](#); [?](#)) shows that financial crises are systematically preceded by large private-credit expansions. Once a crisis hits, households and firms deleverage, reduce consumption and investment, and experience slowed output growth. Similar to previous studies on public debt, [Mian and Sufi \(2010\)](#); [Mian et al. \(2013, 2017\)](#) find that public debt dynamics are endogenous to private-sector distress. They found that private-sector revenue losses lead to government balance-sheet expansion through higher unemployment benefits, bank rescues, and fiscal support. This perspective implies that spikes in public debt are often a consequence of macroeconomic weakness rather than its underlying cause.

The methodological foundation of this paper builds on the flexibility of local projections (LPs) introduced by [Jordà \(2005\)](#) and recent advances integrating LPs with difference-in-differences logic. The LP-DiD framework developed by [Jordà and Taylor \(2023\)](#) shows how local projections can implement a dynamic difference-in-differences design, allowing the researcher to estimate horizon-specific treatment effects even when treatment timing is staggered across units. This is particularly useful in macroeconomic settings, where policies and economic events often propagate across countries over time.

The LP-DiD approach uses never-treated or not-yet-treated units as controls and removes treated observations from the control group once treatment occurs. As a result, the estimator mitigates problems such as treatment contamination and negative weighting that can arise in conventional staggered-treatment DiD or simpler LP specifications. Another key advantage of LP-DiD is that pre-treatment and post-treatment dynamics are estimated within a single regression framework, allowing the researcher to conduct event-study style diagnostics within the same specification. For these reasons, the method is particularly well-suited to macroeconomic settings in which treatment timing may be endogenous, staggered over long periods, and where countries may exhibit substantial heterogeneity.

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Local projections produce horizon-specific regressions that directly estimate the dynamic response to a treatment. This model is often used in macroeconomic research because it is robust to model misspecification and well-suited for heterogeneous or nonlinear dynamics across countries. Individual horizon regressions allow flexible conditioning on covariates and fixed effects, enabling the incorporation of country and time fixed effects, lagged and contemporaneous variables, in a single specification without imposing strong parametric restrictions.

This type of flexibility is essential for studying macroeconomic policies as treatment effects. Changes in policy are not usually randomly assigned treatments. This is especially important when considering debt threshold endogeneity and its effects. Countries may be more likely to cross high public or private debt levels during booms, busts, or crises. LP-DiD provides a transparent way to assess these dynamics by separating observations into groups that have ever been treated versus those never treated in the future. This prevents the usual pitfalls of simpler DiD models by allowing treated groups to remain out of the control group at all future horizons, avoiding contamination that arises when already-treated units re-enter the comparison set.

By combining these advances, this paper contributes to the empirical literature on how

public and private debt spikes relate to macroeconomic outcomes. It also revisits the foundational public-debt literature through a framework that conditions on crisis periods and explicitly tests for pre-trend behavior. With these new advances, this paper aims to reconcile the empirical evidence on the negative relationship between public debt and growth with more recent evidence on their endogeneity, and to connect this perspective to emerging research on private credit booms, crisis dynamics, and persistent output losses.

3. Empirical Methodology

This section describes the local-projection difference-in-differences (LP–DiD) strategy used to estimate the dynamic effects of crossing public- and private-debt thresholds. Countries are observed annually and indexed by i and t , with all specifications estimated on an unbalanced macroeconomic panel.

3.1. Local-Projection DiD Framework

For a given threshold T and horizon $h \geq 0$, the baseline specification is:

$$Y_{i,t+h} = \alpha_i + \lambda_t + \beta_h \text{Shock}_{it}^T + \gamma' X_{it} + \varepsilon_{i,t+h}, \quad (1)$$

where α_i and λ_t denote country and year fixed effects, respectively, and X_{it} is a vector of lagged macro-financial controls. The treatment indicator is coded as a *first-crossing* event:

$$\text{Shock}_{it}^T = 1\{T_{it} = 1 \text{ and } T_{i,t-1} = 0\},$$

ensuring that β_h captures the dynamic adjustment following the initial switch above the threshold, rather than persistent high-debt exposure.

3.2. Horizon Construction

Local-projection horizons $h = 0, \dots, 10$ are constructed to match macroeconomic conventions:

- **Real GDP:** cumulative log deviations from the pre-crossing level:

$$Y_{i,t+h} = 100 [\log(\text{RGDP}_{i,t+h}) - \log(\text{RGDP}_{i,t-1})].$$

- **Other outcomes:** non-cumulative changes in consumption, investment, inflation, unemployment, and credit.

All outcome–threshold–horizon combinations are estimated separately.

3.3. Baseline Controls

The baseline control vector includes lagged measures of macro-financial conditions:

$$X_{it} = \left\{ \begin{array}{l} \text{trade openness}_{i,t-1}, \text{ unemployment}_{i,t-1}, \text{ credit growth}_{i,t-1}, \\ \text{housing returns}_{i,t-1}, \text{ crisis indicator}_{i,t-1}, \log(\text{population}_{it}) \end{array} \right\}.$$

These controls absorb slow-moving macroeconomic forces correlated with both debt dynamics and growth. Summary statistics for the restricted estimation sample appear in Table 3. Unrestricted-sample summaries are reported in Appendix Table B1.

3.4. Crisis-Window Specification

Because debt surges often occur around financial crises, I estimate a variant that restricts treatment to crossings occurring within the first three years after crisis onset:

$$\text{Shock}_{it}^{T, \text{cw}3y} = \text{Shock}_{it}^T \cdot 1\{0 \leq t - \text{CrisisStart}_i \leq 3\}.$$

This design evaluates whether threshold effects are concentrated during periods of financial stress.

3.5. Robust Specification

To ensure results are not driven by omitted financial indicators, I augment the control vector with:

$$X_{it}^{\text{robust}} = X_{it} \cup \left\{ \log(\text{CA}/\text{GDP}_{i,t-1}), \text{ short-term rate}_{it}, \text{ bond yield}_{it} \right\}.$$

Summary statistics for these measures also appear in Table 3.

3.6. Estimation and Inference

All specifications are estimated via fixed effects with standard errors clustered at the country level. Estimated coefficients β_h for each horizon form the impulse-response functions presented in Section 5.

3.7. Identification Checks

Identification relies on two conditions: (i) parallel trends in the pre-crossing period, and (ii) clean, staggered treatment timing.

Parallel Trends Figure 1 compares average Real GDP growth for countries that ever cross the 90% public-debt threshold to countries that never do. The pre-treatment trajectories move closely together, with no persistent divergence. Minor differences in the late 1990s are short-lived and likely reflect regional banking crises (in the Nordic and U.K. regions) rather than systematic pre-trends. With country and year fixed effects, and the JST crisis indicator included in X_{it} , these shocks are absorbed by the model, supporting the plausibility of the parallel-trends assumption.

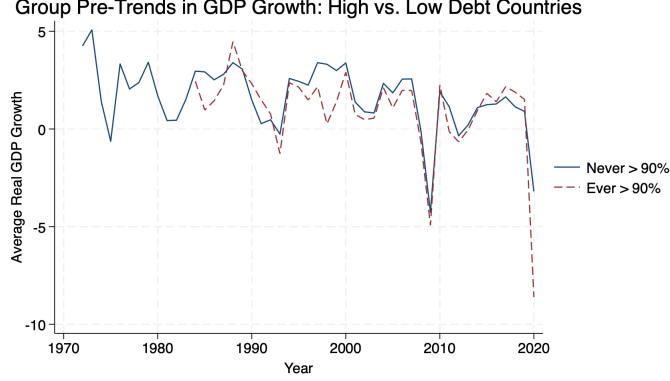


Figure 1: Parallel Trends Test for 90% Public Debt

Clean and Staggered Treatment Timing Each country contributes at most one treatment event per threshold because first-crossings do not reverse. Treatment timing is also staggered across countries, with some crossing in the 1980s and others only after 2008. This structure avoids problems arising from repeated or reversible treatments and eliminates heterogeneous-treatment-timing bias present in traditional two-way DiD estimators. Table 2 documents the staggered timing and number of treated observations in the restricted sample. Unrestricted versions appear in Appendix Table A1.

4. Data

The analysis began with annual macroeconomic data for advanced economies from 1970 to 2020. The primary source is the MacroHistory Database (Jordà et al., 2017, 2021a), which provides harmonized series for output, prices, credit, debt, and financial conditions. Financial crises follow the JST chronology (Jordà et al., 2021b).

To ensure comparability between the baseline control variables, the estimation sample is restricted to observations for which the full set of macro-financial variables and baseline controls is available. This reduces the usable sample from 912 to 769 country-year observations, approximately 84% of the original data. The resulting balanced sample then spans 1972–2020. This is convenient for ensuring stable comparison because it naturally excludes

the Bretton Woods period.

4.1. Outcome Variables

Real GDP growth is the primary outcome, defined as the annual log change in Real GDP. Additional outcomes include consumption growth, investment growth, inflation, credit growth, and changes in unemployment. Summary statistics appear in Table 1.

Table 1: Summary Statistics: Outcomes and Debt Variables (Restricted Sample)

Variable	Count	Mean	Median	SD	Min	Max
Real GDP growth (Δ RGDP)	769	1.73	1.82	2.62	-11.94	20.56
Consumption growth	769	1.57	1.66	2.52	-13.24	10.11
Investment growth	768	7.61	5.87	7.84	-26.32	48.80
Inflation	769	4.02	2.51	4.17	-4.58	32.12
Credit growth	769	7.94	7.43	7.09	-24.83	37.28
Unemployment change	769	0.06	0.00	1.03	-3.44	6.60
Public debt (% GDP)	769	0.57	0.50	0.34	0.04	2.34
Private debt (% GDP)	769	0.85	0.79	0.34	0.20	2.04

4.2. Debt Measures and Thresholds

Public debt is measured as general government gross debt relative to GDP; private debt is bank credit to the non-financial private sector relative to GDP. Thresholds are defined at 60%, 70%, 80%, and 90% for public debt and at 100%, 120%, and 135% for private debt. Restricted-sample diagnostics for crossing events appear in Table 2.

Table 2: Diagnostics for Public and Private Debt Thresholds (Restricted Sample)

Threshold	# Countries	First Crossing Years	Treated Obs
Panel A: Public Debt Thresholds			
Public debt > 60%	22	1982, 1983, 1984 (2), 1985, 1992 (2), 1994 (2), 1995, 1997, 1999, 2002 (2), 2003, 2004, 2009, 2010, 2011, 2015, 2020 (2)	269
Public debt > 70%	11	1984, 1986, 1987, 1993, 1995, 2005, 2009 (3), 2010, 2012	182
Public debt > 80%	9	1986, 1994, 2009 (3), 2010, 2012 (3)	152
Public debt > 90%	8	1990, 1995, 2008, 2010 (2), 2012, 2013, 2020	130
Panel B: Private Debt Thresholds			
Private debt > 100%	16	1974, 1979, 1985, 1989, 1993, 1998, 1999, 2000, 2004 (2), 2005, 2006, 2008, 2014, 2020 (2)	203
Private debt > 120%	9	1986, 1987, 2002, 2005, 2006, 2007, 2015, 2016, 2020	123
Private debt > 135%	9	1988, 2003, 2004, 2006, 2008, 2009, 2012, 2013, 2019	81

Notes: Numbers in parentheses indicate the number of countries whose first crossing occurs in that year. The restricted estimation sample contains 769 country–year observations.

4.3. Control Variables

Summary statistics for the baseline and robustness control variables appear in Table 3. Unrestricted-sample control statistics and correlation matrices appear in Appendix Tables B1–B3.

Table 3: Summary Statistics: Baseline and Robustness Controls (Restricted Sample)

Variable	Count	Mean	Median	SD	Min	Max	% Missing
Lagged trade openness	769	0.56	0.49	0.35	0.11	2.76	0.00
Lagged unemployment rate (%)	769	6.87	6.29	4.00	0.04	26.09	0.00
Lagged log CA/GDP	403	0.78	0.95	1.14	-4.75	2.77	47.6
Lagged credit growth (%)	759	1.00	0.98	4.63	-35.30	20.61	1.3
Lagged real housing return	686	0.11	0.11	0.09	-0.13	0.96	10.8
Short-term interest rate (t)	694	0.05	0.05	0.05	-0.02	0.21	9.7
Bond trading volume (t)	695	0.08	0.07	0.10	-0.31	0.81	9.6
Crisis indicator (t-1)	769	0.03	0.00	0.17	0.00	1.00	0.00
Log population	769	9.91	9.67	1.23	7.99	12.70	0.00

5. Empirical Results

This section presents the dynamic effects of crossing major public and private debt thresholds using the LP-DiD framework introduced above. The analysis proceeds in three steps. First, we examine baseline responses in the full sample, focusing on Real GDP dynamics around key public and private thresholds. Second, we study how these responses differ when conditioning on crisis timing, which can help us distinguish structural effects from possible fiscal or spending adjustments across an economy that may occur in response to an ongoing financial crisis. Finally, we evaluate pre-treatment trends using negative horizons to support the identification of our model and assess the context of threshold timing. Pulled together, these results build a holistic analysis of the differences in how public versus private debt shapes post-crisis macroeconomic performance and macroeconomic performance as a whole.

5.1. Baseline full-sample responses of Real GDP

We begin by documenting the baseline LP-DiD response of Real GDP to crossing the 90% public-debt threshold as our benchmark, as it has long been viewed as such in the literature. Figure 2 reports the estimated dynamic response of Real GDP following the first year in which public debt exceeds 90% of GDP.

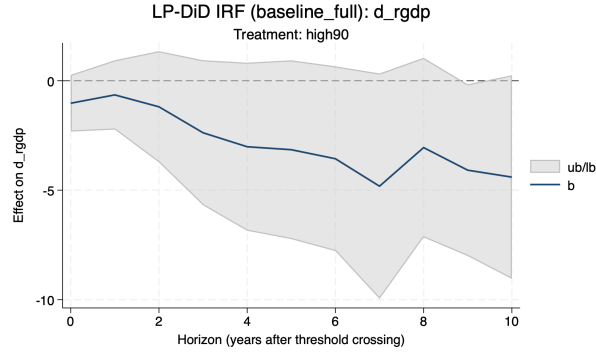


Figure 2: Baseline LP–DiD response of Real GDP to crossing the 90% public-debt threshold.

The point estimates suggest a modest decline in Real GDP growth (hovering between -2% and -1%) in the immediate years after the threshold is crossed, with a sharp decline hovering near 5% around year 7. Yet the response is statistically imprecise. Our confidence intervals include just around 0% or above and remain wide throughout the horizon. This pattern indicates that in our full sample, the 90% threshold does not generate a sharp or well-defined break in Real GDP activity.

To contextualize this finding, Figure 3 compares responses for the 90% public-debt threshold with a lower public threshold at 80% and a high private-debt threshold at 120%. This comparison provides a useful reference for assessing whether the 90% threshold behaves differently from other public thresholds, and whether private leverage dynamics generate more pronounced macroeconomic effects.

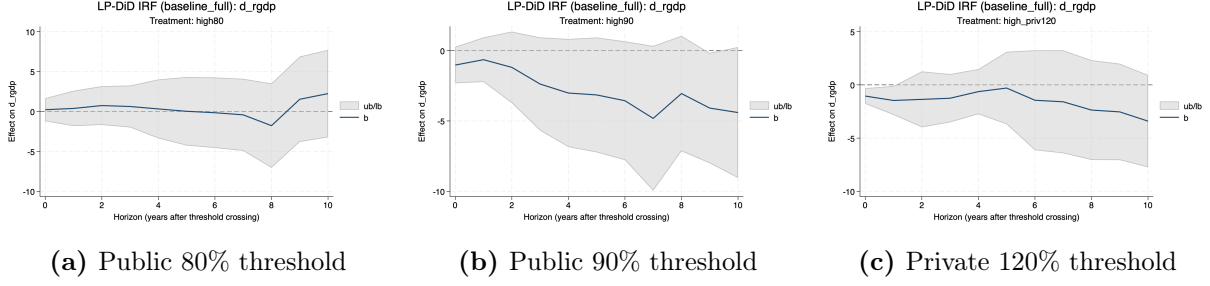


Figure 3: Baseline LP–DiD responses of Real GDP at alternative debt thresholds.

The comparison shows that public-debt thresholds—whether at 80% or 90%—produce mild and imprecise Real GDP effects. At the 80% public debt threshold, estimated impulse responses seem more stable, and differ from those of countries that meet the 90% threshold. In contrast, crossing the private-debt threshold at 120% of GDP is followed by a noticeably clearer decline in Real GDP growth in the immediate years following. By year five, the likelihood of negative output increases, but again provides an imprecise estimate. This difference suggests that private leverage may be a more informative indicator than public debt for identifying emerging macroeconomic weakness, even before conditioning on crisis timing.

5.2. Crisis-window responses and the attenuation of public debt effects

Now we consider our thresholds with the backdrop of a financial crisis. Figure 4 plots, for each public and private threshold, the share of first crossings occurring within three years of a JST-identified crisis.

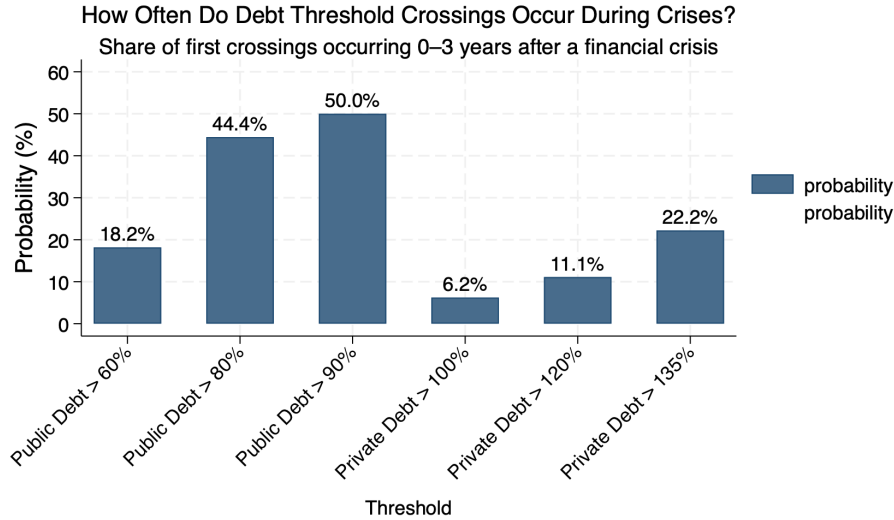


Figure 4: Share of first threshold crossings occurring 0–3 years after a financial crisis.

Crossing the public 80% and 90% thresholds occurs disproportionately during, or immediately after, financial crises. Roughly 44% of 80% crossings and 50% of 90% crossings fall within crisis windows. This is consistent with empirical evidence that severe recessions trigger both automatic stabilizers (welfare or unemployment expansions) and discretionary fiscal responses (bailouts or stimulus packages), leading to a jump in public debt ratios during a recession or crisis. The 90% threshold, which became a focal point in the austerity debate following [Reinhart and Rogoff \(2010\)](#), may need to be interpreted more as a by-product of crises and counter-cyclical spending.

By contrast, private-debt thresholds display a very different pattern. Crossings of 100% and 120% private debt are rarely concentrated in crisis windows (only 6% and 11% respectively). The higher 135% threshold shows slightly more crisis clustering, but still far less than for public-debt thresholds. This is consistent with the logic that private debt expands during booms, and only becomes a crisis when major shocks to banking or credit occur. This distinction supports focusing on a high private threshold, such as 120%, because it captures episodes of high private leverage that are less likely in response to a crisis.

The crisis clustering of 80% and 90% public-debt crossings also illuminates the subdued

or stable Real GDP growth behavior found in our baseline LP-DiD results. Crossing either threshold tends to occur in years when output is already falling. Both for consistency with the literature and to capture episodes of high debt that respond to crises, the analysis that follows emphasizes the 90% threshold.

Because public debt tends to rise sharply during financial crises, it is possible that full-sample estimates confound debt effects with endogenous fiscal responses to economic distress. To address this concern, I estimate a 3-year crisis-window specification in which treatment is restricted to threshold crossings occurring within three years of a financial crisis. This allows a cleaner comparison of the structural implications of public and private debt thresholds.

Figure 5 reports the crisis-window responses associated with crossing the 90% public-debt threshold across four key macroeconomic outcomes: Real GDP, consumption, investment, and unemployment.

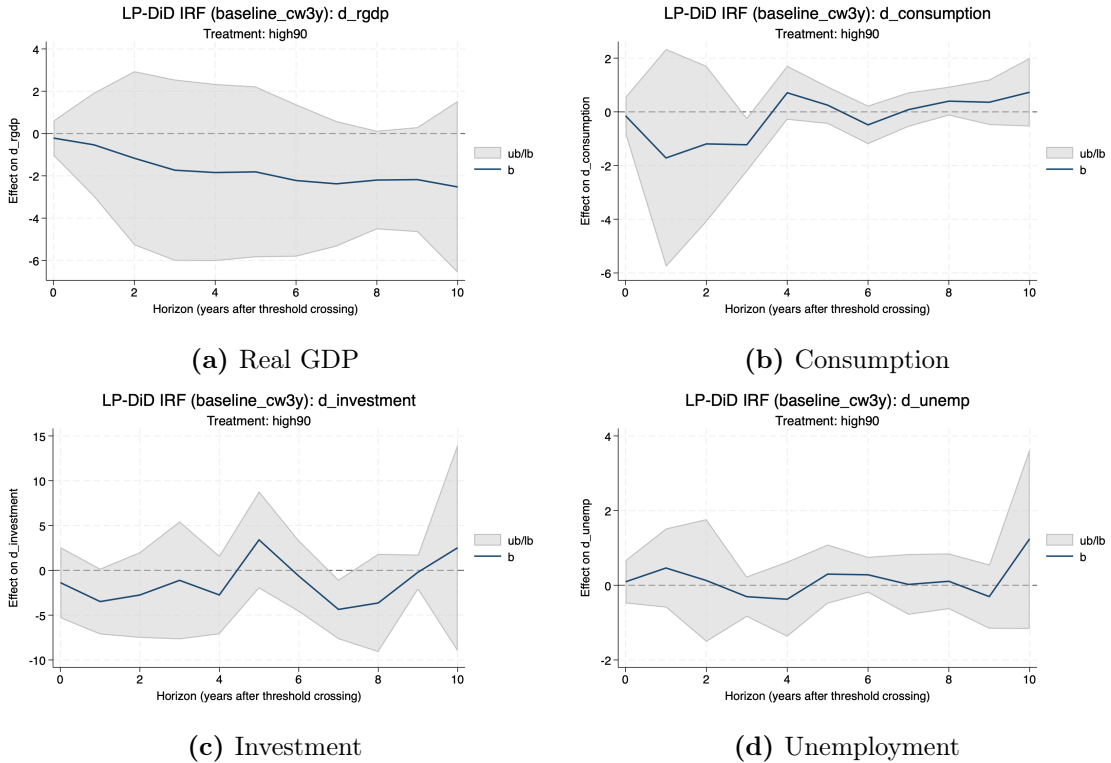


Figure 5: Crisis-window LP-DiD responses for the 90% public-debt threshold.

Once the analysis conditions on crisis timing, the previously negative pattern associated

with the 90% threshold is even less pronounced. The estimated responses for Real GDP, consumption, and investment fluctuate around zero, and the unemployment effects are small and statistically uncertain. This attenuation is consistent with the interpretation that high public debt during crises reflects endogenous counter-cyclical fiscal activity rather than an obstacle to economic growth.

In contrast, Figure 6 shows that the private 120% debt threshold continues to produce pronounced, notably negative responses across Real GDP, consumption, investment, and unemployment, even within crisis windows. Our confidence intervals are especially tight in our consumption and unemployment changes.

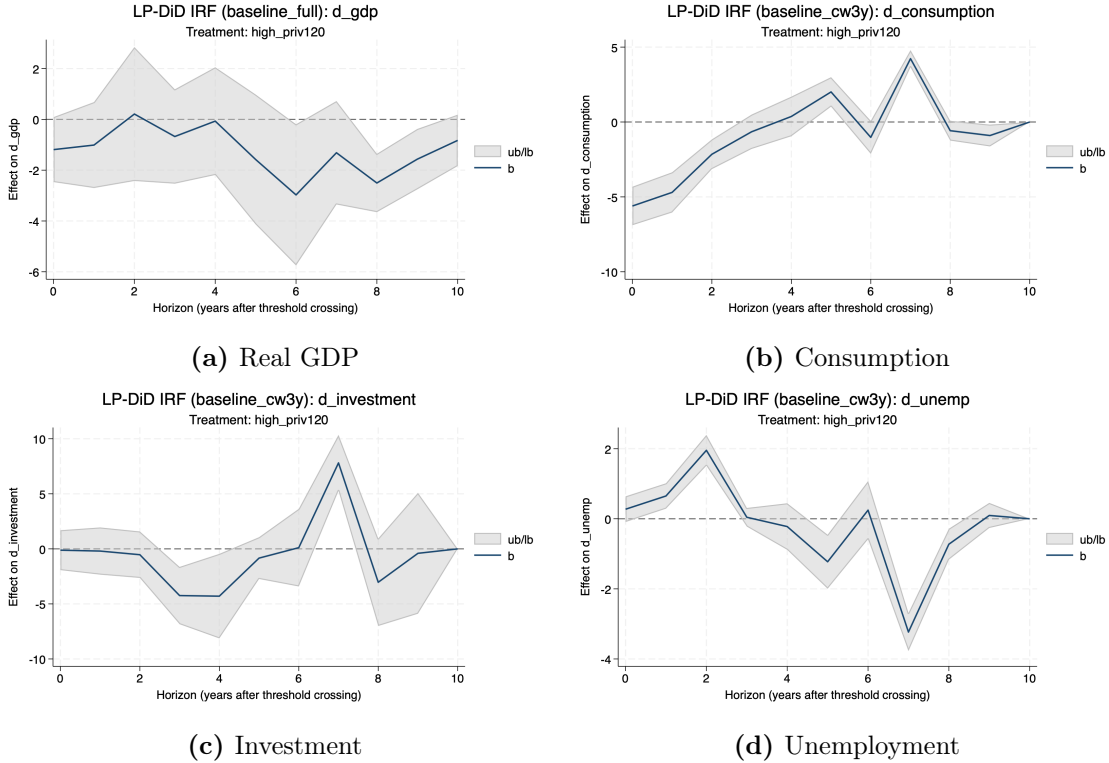


Figure 6: Crisis-window LP-DiD responses for the 120% private-debt threshold.

Not only do we see a sharp decrease in consumption at about 5% immediately following the threshold crossing, but we also see a 2% spike in unemployment building in the 2 years preceding the threshold crossing. We also see a more persistent negative trend for all outcomes. These findings reinforce the view that private leverage is a better proxy for analyzing

structural macro-financial fragility.

5.3. Pre-trend evidence: public versus private debt thresholds

To evaluate identification more formally, I extend the LP-DiD framework to include negative horizons to provide a more direct source of evidence on the pre-treatment environment. This approach is similar to an event-study design, where we can see if countries crossing the threshold were already on differential trajectories prior to treatment.

Figure 7 displays our side-by-side pre-trend graphs for Real GDP for the public 80% and 90% thresholds. Both thresholds exhibit declines in Real GDP growth in the years leading up to the treatment crossing. Consistent with our baseline results and probabilities of crisis occurrence, we see our 90% threshold countries experience a slightly stronger decline in Real GDP than our 80% threshold countries. Once again, these observations complicate a causal interpretation of the impact of public debt thresholds on Real GDP growth.

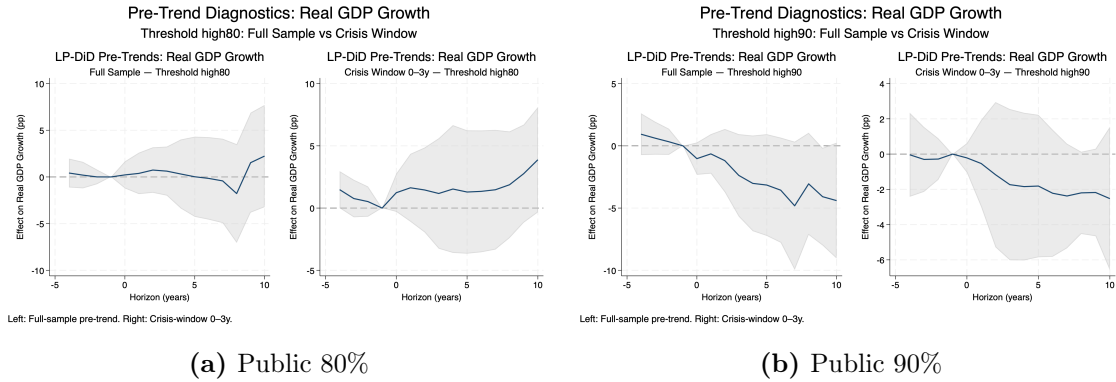


Figure 7: Pre-trend diagnostics for Real GDP around public-debt thresholds.

Figure 8 shows that in the full sample (left panel (b)), Real GDP growth is nearly stable around 0 in the years leading up to the 120% private-debt threshold before dropping in the immediate years after the crossing. This is sharply contrasted by our crisis-window specification (right panel (b)), where growth is significantly negative ahead of the threshold, and falls even deeper after the crossing. This pattern coincides with the view that rapid credit expansion during booms may create greater exposure or vulnerabilities, making it a

more reliable early sign of macroeconomic downturns.

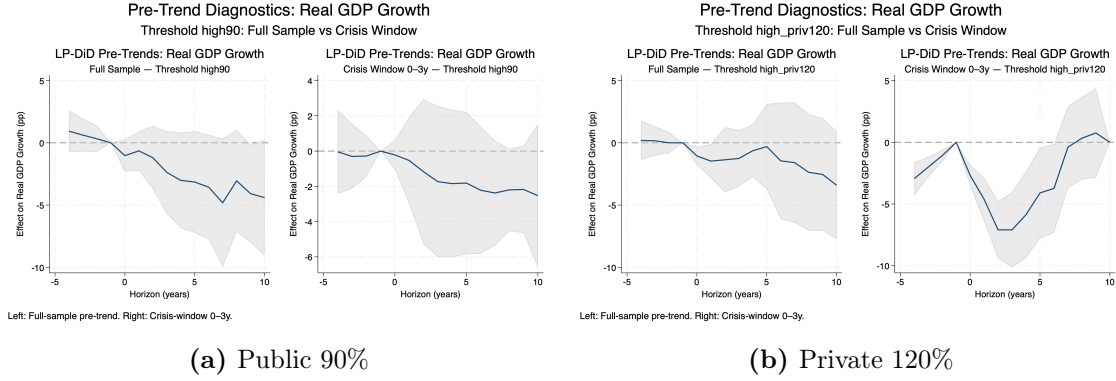


Figure 8: Pre-trend comparison for RGDP: public vs. private thresholds.

Consumption trends reinforce these ideas. As shown in Figure 9, consumption growth declines moderately before crossing the 90% public threshold, but is more pronounced for the private 120% threshold. During a crisis, high credit thresholds can offer more significant estimates than outside of a crisis, but this could also be because credit booms are much less likely to occur during a crisis. These results are consistent with theories of private sector deleveraging.

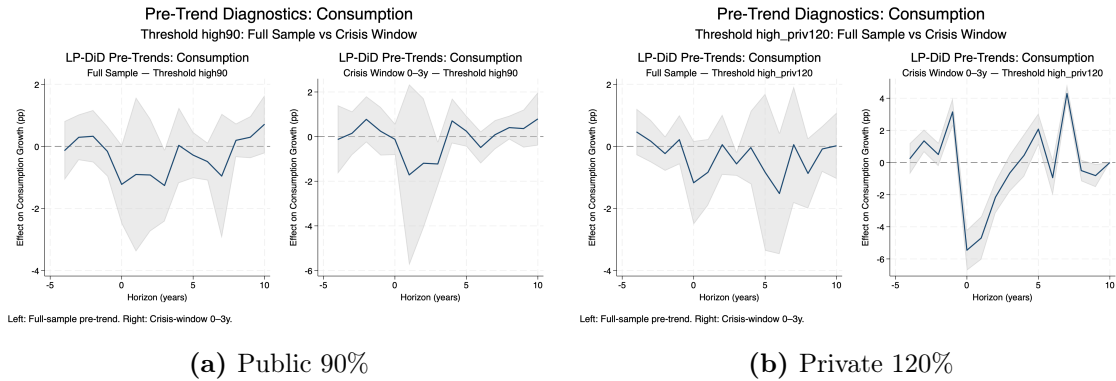


Figure 9: Pre-trend comparison for consumption: public vs. private thresholds.

Overall, the pre-trend evidence shows that the negative associations often attributed to high public debt tend to occur during periods in which economies are already facing growth contraction. Therefore, interpreting macroeconomic downturns through public debt accumulation faces significant obstacles to disproving its endogeneity in support of causal inference.

On the other hand, economies with exceptionally high private credit, especially when these episodes coincide with crises, exhibit significantly worse macroeconomic conditions both before and after the threshold is crossed. This pattern aligns with the literature that suggests private leverage amplifies downturns and that episodes of credit booms can be linked to the severity of business cycles.

6. Conclusion

This paper re-examined the relationship between high public and private debt thresholds across countries and their macroeconomic performance using LP-DiD estimation. The empirical evidence reveals a sharp contrast between the behavior of public debt thresholds and private debt thresholds, with important implications for how researchers interpret high-debt episodes. Across the full sample, apparent negative responses of Real GDP growth following the public 90% debt threshold initially resemble the “threshold effects” emphasized in early empirical work. Real GDP growth and consumption exhibit a downward trajectory in the five years preceding the threshold crossing. When the sample is further restricted to crossings that occur within three years of a financial crisis, these pre-trends virtually disappear while post-treatment coefficients fluctuate around zero. This interpretation is reinforced by the fact that half of all 90

High private debt thresholds are associated with a more negative pattern in macroeconomic outcomes. Crossing the 120% private debt threshold is associated with deeper and sustained declines in Real GDP growth. The pre-treatment dynamic path shows relatively stable output before a country reaches the 120% threshold, followed by a sharp fall once the threshold has been exceeded. Consumption responds even more pronouncedly. In the 3-year crisis-window sample, consumption tumbles in the immediate period as a country reaches its debt threshold, consistent with the deleveraging documented in macro-finance literature. Unemployment spikes just after the private threshold is crossed. Also consistent with research that suggests private-sector spillovers could be closely linked to spikes in

private debt.

Altogether, the results suggest that public and private debt thresholds can reveal a more holistic view of macroeconomic mechanisms. Public debt thresholds can reflect the fiscal impacts of crises and recessions, while private-debt thresholds are more closely tied to a country’s structural vulnerabilities in the credit cycle. This evidence may motivate more research into how high private leverage precedes and amplifies downturns. The LP-DiD framework proves invaluable for uncovering these distinctions across time and business cycles. By allowing one to estimate pre-trends and the dynamic effects of treatments, one can more transparently assess causal relationships between macroeconomic outcomes and staggered policy or event treatment variables.

Future research could extend these findings by studying joint public–private debt interactions, exploring heterogeneous country responses based on structural differences in credit access, or by incorporating forward-looking indicators of private credit conditions. More granular data around household, firm, and government spending within a country across crisis events could also illuminate how a country’s economy may be more vulnerable to cyclical risks.

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Appendix A: Data Definitions and Construction

A.1 Variable Descriptions

A.1 Variable Descriptions

All macroeconomic and financial variables are taken from the JST MacroFinance & MacroHistory Database (Jordà et al., 2017, 2021a), which provides harmonized annual data on real and nominal GDP, consumption, investment, prices, employment, credit, debt, and external balances. I adopt the JST variable definitions throughout and depart from them only when constructing growth rates, ratios, or indicator variables as described below.

A.1.1 Outcome Variables

All outcomes are expressed as annual changes. Let L denote the one-year lag and $\log(\cdot)$ the natural logarithm. For each country i and year t :

$$\text{Real GDP growth: } d_{\text{rgdp},it} = 100[\log(\text{RGDP}_{it}) - \log(\text{RGDP}_{i,t-1})],$$

$$\text{Nominal GDP growth: } d_{\text{gdp},it} = 100[\log(\text{NGDP}_{it}) - \log(\text{NGDP}_{i,t-1})],$$

$$\text{Real GDP per capita growth: } d_{\text{rgdppc},it} = 100[\log(\text{RGDP}_{it}/\text{POP}_{it}) - \log(\text{RGDP}_{i,t-1}/\text{POP}_{i,t-1})],$$

$$\text{Consumption growth: } d_{\text{cons},it} = 100[\log(\text{CONS}_{it}) - \log(\text{CONS}_{i,t-1})],$$

$$\text{Investment growth: } d_{\text{inv},it} = 100[\log(\text{INV}_{it}) - \log(\text{INV}_{i,t-1})],$$

$$\text{Inflation: } \inf_{it} = 100[\log(\text{CPI}_{it}) - \log(\text{CPI}_{i,t-1})],$$

$$\text{Credit growth: } d_{\text{cred},it} = 100[\log(\text{CREDIT}_{it}) - \log(\text{CREDIT}_{i,t-1})],$$

$$\text{Change in unemployment: } d_{\text{unemp},it} = \text{UNEMP}_{it} - \text{UNEMP}_{i,t-1}.$$

All log differences are scaled by 100 so that values are in percentage points. The first observation for each country is omitted from each constructed series.

A.1.2 Debt Ratios and Threshold Indicators

Public debt is measured as general government gross debt divided by nominal GDP. Private debt is banking-sector credit to the non-financial private sector divided by nominal GDP. Threshold indicators are defined as:

$$\text{PublicDebt}_{it} > \{60\%, 70\%, 80\%, 90\%\}, \quad \text{PrivateDebt}_{it} > \{100\%, 120\%, 135\%\},$$

where each indicator equals one in the year the ratio first exceeds the cutoff. These treatment definitions follow the LP–DiD framework used in the main text.

A.1.3 Crisis Indicator

Financial crisis dates follow the JST crisis chronology ([Jordà et al., 2021b](#)). The variable Crisis_{it} is equal to one in crisis years. A crisis-window dummy equals one in the crisis year and the subsequent three years, capturing the macro-financial environment in which many public-debt thresholds are crossed.

A.1.4 Control Variables

All controls are lagged one year in the LP–DiD specification:

- Trade openness: exports + imports divided by GDP. - Unemployment rate. - Log current-account balance scaled by GDP. - Real credit growth. - Real housing returns. - Short-term money-market interest rate. - Long-term government bond market turnover. - Log population (captures scale and openness effects).

A.2 Correlation Matrix and Diagnostics

To assess the relationship among the macro-financial controls used in the baseline and robustness specifications, [Table 4](#) reports the pairwise correlation matrix for the restricted

sample ($N = 403$).

Table 4: Correlation Matrix of Control Variables (Restricted Sample, $N = 403$)

	Trade Open	Unemp	Log(CA/GDP)	Credit Gr.	Housing Ret.	Bill Rate	Bond Vol.	Crisis (t-1)	Log Pop
Trade openness	1.00	0.17	0.20	-0.03	0.02	-0.20	-0.02	0.01	-0.39
Unemployment	0.17	1.00	-0.14	-0.38	-0.15	0.02	0.22	-0.08	0.09
Log(CA/GDP)	0.20	-0.14	1.00	0.09	-0.12	-0.44	-0.09	0.05	-0.35
Credit growth	-0.03	-0.38	0.09	1.00	0.12	0.06	-0.19	0.05	-0.10
Housing return	0.02	-0.15	-0.12	0.12	1.00	0.42	-0.02	-0.10	-0.19
Short-term rate	-0.20	0.02	-0.44	0.06	0.42	1.00	0.25	-0.08	0.08
Bond volume	-0.02	0.22	-0.09	-0.19	-0.02	0.25	1.00	-0.06	-0.01
Crisis (t-1)	0.01	-0.08	0.05	0.05	-0.10	-0.08	-0.06	1.00	-0.01
Log population	-0.39	0.09	-0.35	-0.10	-0.19	0.08	-0.01	-0.01	1.00

Appendix B: Supplementary Tables

B.1 Summary Statistics (Full Sample)

Table 5: Summary Statistics: Baseline and Robustness Controls (Full Sample)

Variable	Count	Mean	Median	SD	Min	Max	% Missing
Lagged trade openness	892	0.57	0.49	0.36	0.11	2.76	2.19
Lagged unemployment rate (%)	892	6.90	6.32	4.04	0.04	26.09	2.19
Lagged log CA/GDP	448	0.79	0.97	1.15	-4.75	2.77	50.88
Lagged credit growth (%)	872	1.01	0.99	4.67	-35.30	20.61	4.39
Lagged real housing return	786	0.11	0.11	0.09	-0.13	0.96	13.82
Short-term interest rate (t)	809	0.05	0.05	0.05	-0.02	0.21	11.29
Bond trading volume (t)	810	0.08	0.07	0.10	-0.31	0.81	11.18
Crisis indicator (t-1)	892	0.03	0.00	0.17	0.00	1.00	2.19
Log population	912	9.93	9.69	1.23	7.99	12.70	0.00

Table 6: Summary Statistics: Treatment Indicators (Full Sample)

Treatment Variable	Count	Mean	SD	Max
Public debt > 60%	912	0.38	0.48	1
Public debt > 70%	912	0.27	0.44	1
Public debt > 80%	912	0.21	0.41	1
Public debt > 90%	912	0.17	0.38	1
Private debt > 100%	912	0.24	0.43	1
Private debt > 120%	912	0.14	0.35	1
Private debt > 135%	912	0.09	0.29	1

Table 7: Summary Statistics: Outcomes and Debt Variables (Full Sample)

Variable	Count	Mean	Median	SD	Min	Max
Real GDP growth (Δ RGDP)	892	1.69	1.79	2.58	-11.94	21.46
Nominal GDP growth	892	6.66	5.54	5.52	-10.36	29.87
Real GDP per capita growth	892	1.08	1.23	2.72	-19.02	20.56
Consumption growth	892	1.54	1.62	2.55	-13.24	10.11
Investment growth	891	7.48	5.74	7.71	-26.32	48.80
Inflation	892	4.29	2.63	4.46	-4.58	32.12
Credit growth	892	8.14	7.54	7.30	-24.83	37.28
Change in unemployment	892	0.08	0.00	1.07	-3.44	6.60
Public debt (% GDP)	912	0.58	0.51	0.35	0.04	2.34
Private debt (% GDP)	912	0.83	0.77	0.33	0.20	2.04

B.2 Threshold Diagnostics (Restricted Sample)

Table 8: Diagnostics for Public and Private Debt Thresholds (Restricted Sample)

Threshold	# Countries	First Crossing Years	Treated Obs
Panel A: Public Debt Thresholds			
Public debt > 60%	22	1982, 1983, 1984 (2), 1985, 1992 (2), 1994 (2), 1995, 1997, 1999, 2002 (2), 2003, 2004, 2009, 2010, 2011, 2015, 2020 (2)	269
Public debt > 70%	11	1984, 1986, 1987, 1993, 1995, 2005, 2009 (3), 2010, 2012	182
Public debt > 80%	9	1986, 1994, 2009 (3), 2010, 2012 (3)	152
Public debt > 90%	8	1990, 1995, 2008, 2010 (2), 2012, 2013, 2020	130
Panel B: Private Debt Thresholds			
Private debt > 100%	16	1974, 1979, 1985, 1989, 1993, 1998, 1999, 2000, 2004 (2), 2005, 2006, 2008, 2014, 2020 (2)	203
Private debt > 120%	9	1986, 1987, 2002, 2005, 2006, 2007, 2015, 2016, 2020	123
Private debt > 135%	9	1988, 2003, 2004, 2006, 2008, 2009, 2012, 2013, 2019	81
Private debt > 140%	12	1989, 1991, 1995, 1999, 2004, 2005, 2006, 2009 (2), 2012, 2014, 2020	59
Private debt > 150%	7	2005, 2006, 2009, 2014, 2015, 2017, 2019	34

Table 9: Summary Statistics: Treatment Indicators (Restricted Sample)

Treatment Variable	Mean	SD
Public debt > 60%	0.35	0.48
Public debt > 70%	0.24	0.43
Public debt > 80%	0.20	0.40
Public debt > 90%	0.17	0.38
Private debt > 100%	0.26	0.44
Private debt > 120%	0.16	0.37
Private debt > 135%	0.11	0.31
Total Observations	769	

Table 10: Correlation Matrix of Control Variables (Restricted Sample, $N = 403$)

	Trade Open	Unemp	Log(CA/GDP)	Credit Gr.	Housing Ret.	Bill Rate	Bond Vol.	Crisis (t-1)	Log Pop
Trade openness	1.00	0.17	0.20	-0.03	0.02	-0.20	-0.02	0.01	-0.39
Unemployment	0.17	1.00	-0.14	-0.38	-0.15	0.02	0.22	-0.08	0.09
Log(CA/GDP)	0.20	-0.14	1.00	0.09	-0.12	-0.44	-0.09	0.05	-0.35
Credit growth	-0.03	-0.38	0.09	1.00	0.12	0.06	-0.19	0.05	-0.10
Housing return	0.02	-0.15	-0.12	0.12	1.00	0.42	-0.02	-0.10	-0.19
Short-term rate	-0.20	0.02	-0.44	0.06	0.42	1.00	0.25	-0.08	0.08
Bond volume	-0.02	0.22	-0.09	-0.19	-0.02	0.25	1.00	-0.06	-0.01
Crisis (t-1)	0.01	-0.08	0.05	0.05	-0.10	-0.08	-0.06	1.00	-0.01
Log population	-0.39	0.09	-0.35	-0.10	-0.19	0.08	-0.01	-0.01	1.00

Discussion of Correlations and Multicollinearity. Some of the robust controls—most notably the current account (CA) balance and the short-term interest rate—exhibit relatively high correlation with one another. This relationship is consistent with [Obstfeld and Rogoff \(1996a, 2003\)](#), who show that persistent CA surpluses tend to arise in economies with low interest rates, high savings, and strong net foreign-asset positions. The CA balance is also strongly correlated with log population, in line with [Aizenman and Pinto \(2005\)](#), who doc-

ument that the volatility of CA/GDP declines with country size.

By contrast, the baseline controls do not display the same level of collinearity, though log population remains negatively correlated with trade openness (-0.39), which reflects the fact that larger economies are typically less trade-dependent (?).

Variance-inflation factors (VIFs) provide further reassurance. In a regression of ΔRGDP on the full set of controls, no variable exhibits a VIF above 1.70—well below standard thresholds for moderate (5) or severe (10) multicollinearity. The mean VIF is 1.33, indicating only modest variance inflation. Because the observed correlations are well understood in macroeconomic theory and do not generate problematic VIFs, we proceed with LP-DiD estimation using the complete control set.

B.3 Crisis-Window Threshold Probabilities

Table 11: Debt Threshold Crossings and Crisis-Window Probabilities

Panel A: Public Debt Thresholds			
Threshold	Total Crossings	Crisis-Window Crossings	Probability (%)
Public debt > 60%	22	4	18.18
Public debt > 80%	9	4	44.44
Public debt > 90%	8	4	50.00

Panel B: Private Debt Thresholds			
Threshold	Total Crossings	Crisis-Window Crossings	Probability (%)
Private debt > 100%	16	1	6.25
Private debt > 120%	9	1	11.11
Private debt > 135%	9	2	22.22

Crisis-window defined as 0–3 years after a financial crisis onset.

Crisis-window crossings occur within 0–3 years after the onset of a financial crisis. The probability column reports the share of all crossings falling inside the window.

Appendix C: Additional Figures

C.1 Debt Distributions and Joint Properties

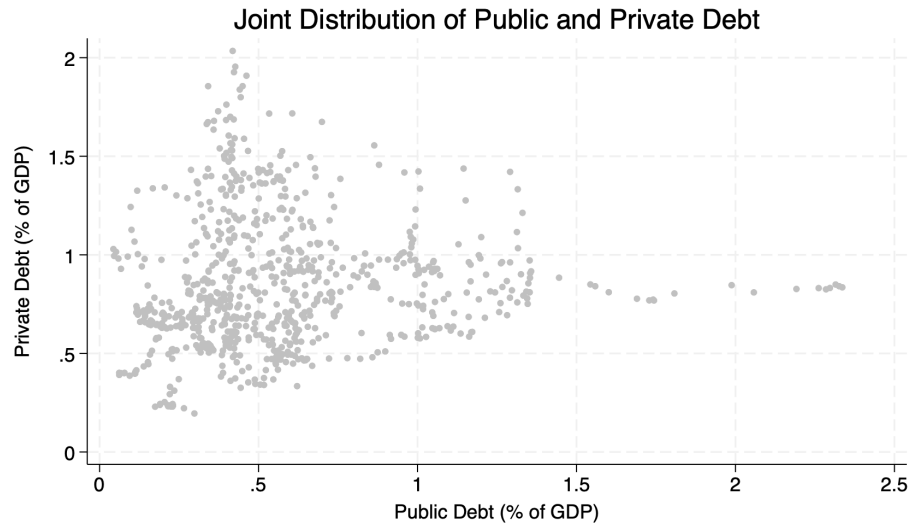


Figure 10: Joint Distribution of Public and Private Debt

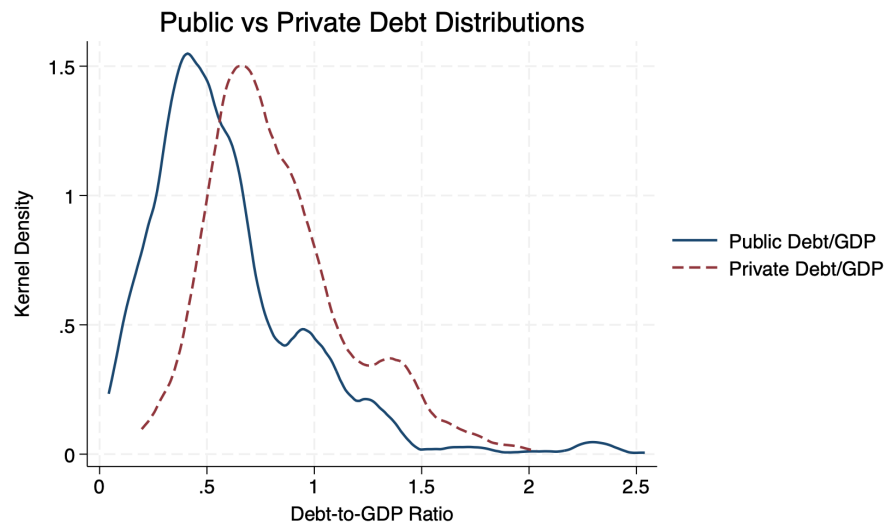


Figure 11: Public vs. Private Debt Distributions

Appendix D: Robustness Checks

This appendix presents robustness results obtained by augmenting the baseline control vector with additional macro-financial indicators (current-account balances, short-term interest rates, long-term bond yields).

Each figure reports impulse responses for Real GDP, consumption, investment, and unemployment in 2×2 grids.

D.1 Public Debt Threshold: 80%

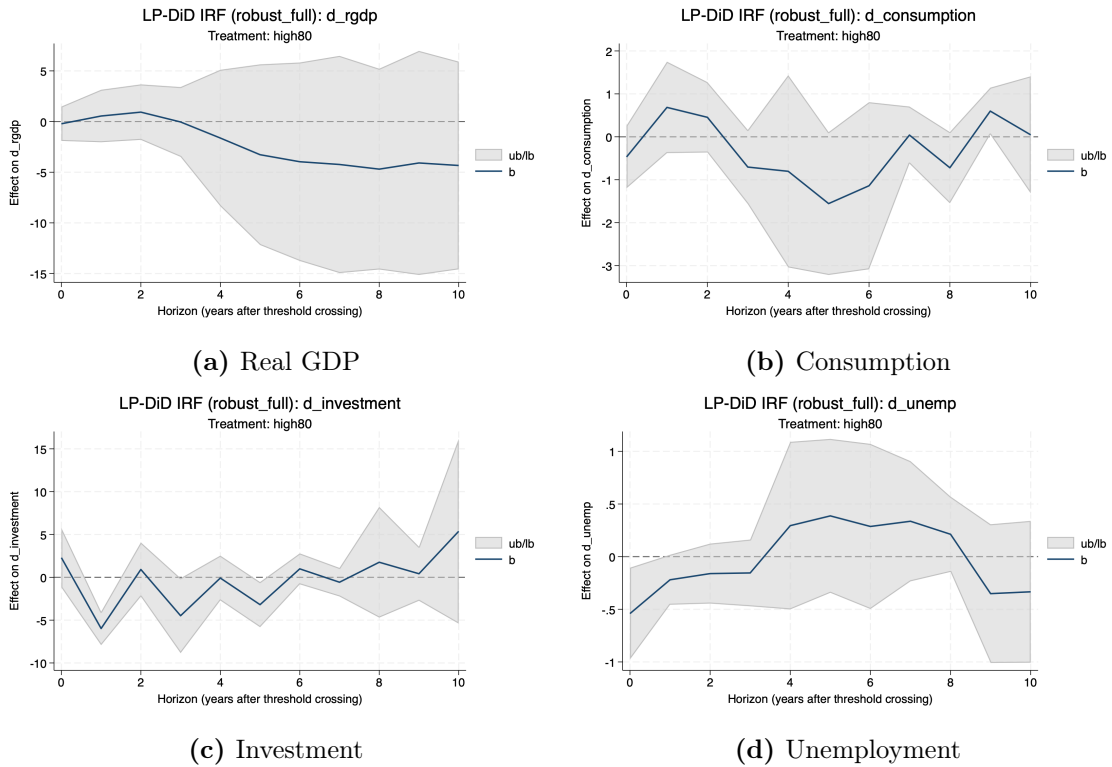


Figure 12: Robust LP-DiD impulse responses for the public 80% debt threshold.

D.2 Public Debt Threshold: 90%

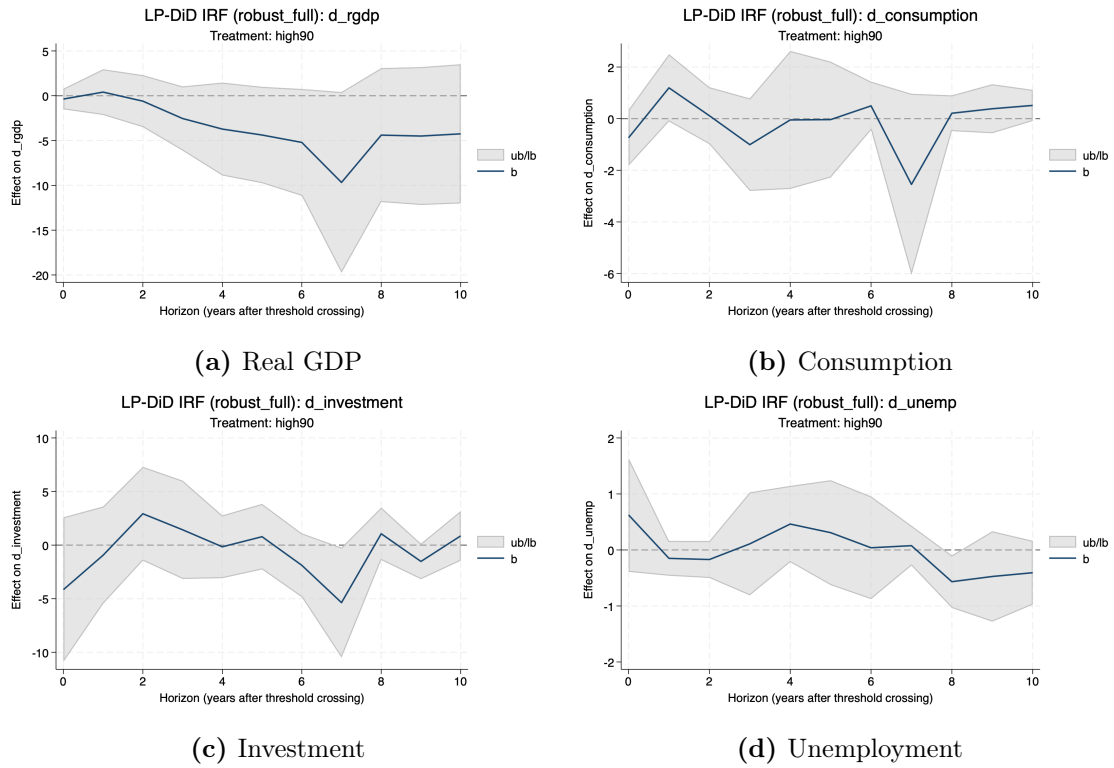


Figure 13: Robust LP-DiD impulse responses for the public 90% debt threshold.

D.3 Private Debt Threshold: 120%

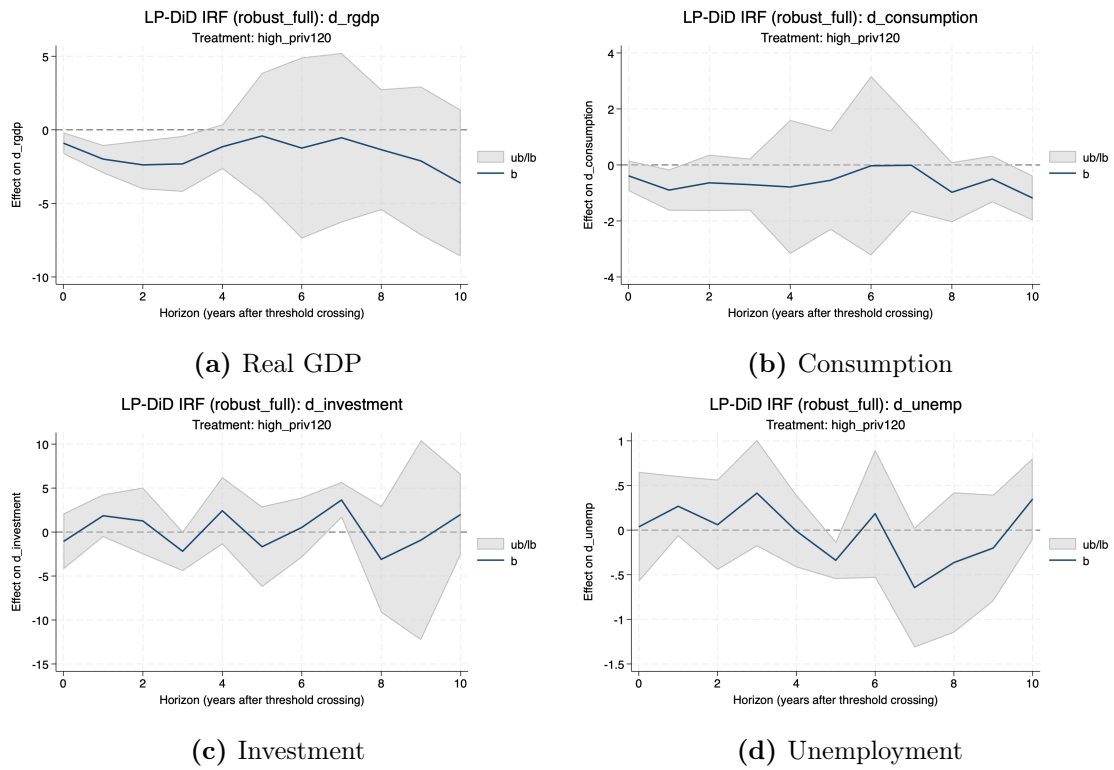


Figure 14: Robust LP-DiD impulse responses for the private 120% debt threshold.